Design and Simulation of 10-Bit SAR ADC Using 18nm FinFET Technology

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***Abstract*—This paper presents a cutting-edge 1.2V 10-bit asynchronous Successive Approximation Register (SAR) Analog-to-Digital Converter (ADC) meticulously designed and simulated in the advanced 18 nm FinFET CMOS technology. This asynchronous SAR ADC system consists of an internal-clock generator, switching unit, fully differential pair of capacitive Digital-to-Analog Converter (DAC), a dynamic comparator, and a SAR logic. The core achievement of this design is evident when considering its practical performance. Employing a 1024-point Fast Fourier Transform (FFT) on the SAR ADC output and enabling a full-scale input range of 0.96 V differentially. At a rapid 100 MHz sampling frequency, this system attains an Effective Number of Bits (ENOB) of 9.7, showcasing the converter's precision and accuracy. Notably, this achievement is complemented by a Signal-to-Noise and Distortion (SINAD) ratio reaching an impressive 60.17 dB, along with a Signal-to-Noise Ratio (SNR) of 61.25 dB, demonstrating the exceptional signal fidelity captured by the ADC. Moreover, the Integral Non-Linearity (INL) and Differential Non-Linearity (DNL) figures are maintained within a remarkable 1-bit range, highlighting the meticulous attention given to achieving linearity and minimizing errors. This ADC system also drawing a total power consumption of 0.216 mW. This SAR ADC system can be used in systems that mainly require high resolution and high speed.**

Keywords—ADC, Successive Approximation Register ADC, high resolution, fully differential pair, FinFET CMOS technology.

I. Introduction

The recent landscape of Analog-to-Digital Converter (ADC) IC design has witnessed remarkable advancements fueled by the demand for higher performance and energy-efficient solutions. A significant focus has been on exploring novel architectures, including Successive Approximation Register (SAR), Delta-Sigma, and Pipeline ADCs [1]. Semiconductor technology has played a pivotal role in pushing the boundaries of ADC design. Advanced processes like FinFET and sophisticated CMOS technologies have enabled the realization of ADCs with improved resolution, reduced power consumption, and superior signal-to-noise ratios [2]. The pursuit of higher sampling rates has become increasingly prevalent, driven by the demands of emerging technologies such as 5G communication and high-speed data acquisition systems [3].

Based on research by [4], millimeter wave technology for 5G networks requires converters operating at GS/s (billion samples per second), achievable only through time-interleaved ADC architectures. The fully differential SAR ADCs proposed in [5] and [6] have drawbacks of design complexity as well as stringent design requirements on the analog circuitry. Thus, a high sampling rate full differential SAR ADC design with reduced complexity and relaxed requirement on comparator is proposed in this paper.

The proposed SAR ADC is using the charge redistribution architecture where capacitive DAC is used to perform both sample and hold operations and digital-to-analog conversion. The analog input is sampled and saved in the capacitor array for each conversion cycle. The output of the DAC is then compared for N clock cycles to a common mode voltage. The common mode voltage at the comparator's input is gradually tracked by the DAC's output during this process, ultimately reaching common mode voltage at the end of each conversion. Because the capacitive DAC has built-in sample-and-hold functionality, one significant benefit of this design is its low power consumption.

II. ADC Architecture

The block diagram and architecture of the proposed 10-bits SAR ADC is shown in Fig. 1 and Fig. 2. It consists of a dynamic comparator, switching capacitive DAC (CAPDAC), and SAR (Successive Approximation Register) circuitry.

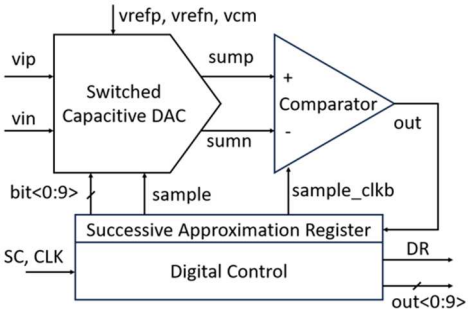


Fig. 1. The proposed SAR ADC block diagram

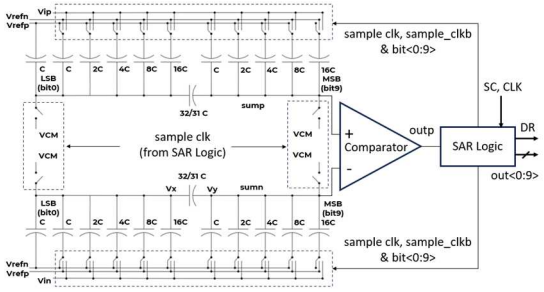


Fig. 2. The proposed SAR ADC architecture

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The differential structure of CAPDAC was utilized in the suggested SAR ADC, and the positive and negative reference voltages were set as given in the equation below:

$$V_{refn} = V_{CM} - \frac{V_{ref}}{2},\ V_{refp} = V_{CM} + \frac{V_{ref}}{2} \quad (1)$$

Improved common-mode rejection ratio (CMRR), decreased even harmonic distortion, and increased conversion accuracy are just a few benefits of using CAPDAC's differential structure. However, because most of the components are doubled for the differential configuration, a larger layout area is required. The three phases of the traditional capacitor switching method (sampling, comparison, and data ready) are used by the SAR ADC [7].

The top half of the CAPDAC differential structure functions in precisely the opposite way as the bottom half. On the lower half, the bottom plates of the capacitors are connected to the negative input voltage (V_{in}) during the sampling phase, while the top plates of the capacitors are connected to the common mode voltage (V_{CM}).

The MSB capacitor's bottom plate will be linked to the negative reference voltage (V_{refn}) during the comparison phase. The charge distribution will take place in the CAPDAC, and the comparator will receive the output (sumn). The comparator output signal becomes high if the input of the comparator sump>sumn, which indicates that the positive input voltage, V_{ip}, is greater than the negative input voltage, V_{in}. As a result, bit 9's value (the MSB capacitor attached to V_{refn}) stays high. Bit 9 (the MSB capacitor attached to V_{refp}) shifts to low if the comparator's input sump<sumn. The ADC proceeds in this manner for every bit, switching between the MSB and LSB, until the value of bit 0 on the LSB is ascertained.

When sampling phase, the charge Q_x at node V_x will be:

$$Q_x = 31C(V_{CM} - V_{in}) + C(V_{CM} - V_{refp}) \quad (2)$$

While, the charge Q_y at node V_y will be:

$$Q_y = 31C(V_{CM} - V_{in}) \quad (3)$$

In the comparison phase, it was assumed that M unit capacitors were connected to V_{refn} and N unit capacitors were connected to V_{refp}. At this time, the charge Q_x at the node V_x could be expressed as follows:

$$Q_x = MC(V_x - V_{refn}) + NC(V_x - V_{refp}) + \frac{32}{31}C(V_x - V_y) + C(V_x - V_{refn}) \quad (4)$$

While, the charge Q_y at node V_y will be:

$$Q_y = MC(V_y - V_{refn}) + NC(V_y - V_{refp}) + \frac{32}{31}C(V_y - V_x) \quad (5)$$

In the process of charge distribution, the charges Q_x and Q_y at nodes V_x and V_y will remain unchanged. Therefore, from the equation above: $Q_x = Q_x$ and $Q_y = Q_y$ [8].

The proposed ADC's successive-approximation approach operation flow is depicted in Fig. 3.

Cycle	Sample	Sample _clkb	DR	bit										comp
				9	8	7	6	5	4	3	2	1	0	out
1	1	0	0	0	0	0	0	0	0	0	0	0	0	-
2	0	1	0	1	0	0	0	0	0	0	0	0	0	a_9
3	0	1	0	a_9	1	0	0	0	0	0	0	0	0	a_8
4	0	1	0	a_9	a_8	1	0	0	0	0	0	0	0	a_7
5	0	1	0	a_9	a_8	a_7	1	0	0	0	0	0	0	a_6
6	0	1	0	a_9	a_8	a_7	a_6	1	0	0	0	0	0	a_5
7	0	1	0	a_9	a_8	a_7	a_6	a_5	1	0	0	0	0	a_4
8	0	1	0	a_9	a_8	a_7	a_6	a_5	a_4	1	0	0	0	a_3
9	0	1	0	a_9	a_8	a_7	a_6	a_5	a_4	a_3	1	0	0	a_2
10	0	1	0	a_9	a_8	a_7	a_6	a_5	a_4	a_3	a_2	1	0	a_1
11	0	1	0	a_9	a_8	a_7	a_6	a_5	a_4	a_3	a_2	a_1	1	a_0
12	0	0	1	a_9	a_8	a_7	a_6	a_5	a_4	a_3	a_2	a_1	a_0	-

Fig. 3. Operation flow of the proposed SAR ADC

III. Circuit Design

Fig. 4 displays the suggested SAR ADC's schematic diagram. The architecture used in the sub-block includes SAR Logic (Verilog-A), CAPDAC (fully differential architecture of split capacitor array DAC), comparator (dynamic comparator using StrongArm latch design) and internal clock generator.

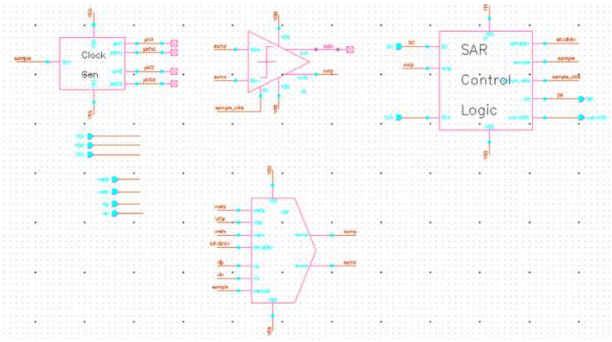


Fig. 4. Proposed SAR ADC schematic diagram

A. SAR Logic

Asynchronous binary search SAR logic architecture is used in the built SAR ADC. Verilog-A is used to create the SAR logic, which models the behaviour of analog circuits utilizing an external clock as an input. The SAR logic schemetic diagram is displayed in Fig. 5. Fig. 6 shows the timing sequence of the SAR logic.

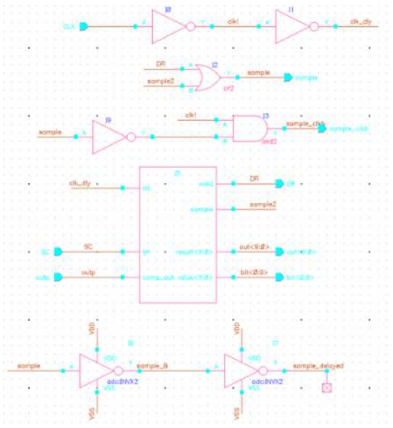


Fig. 5. Schematic of SAR logic

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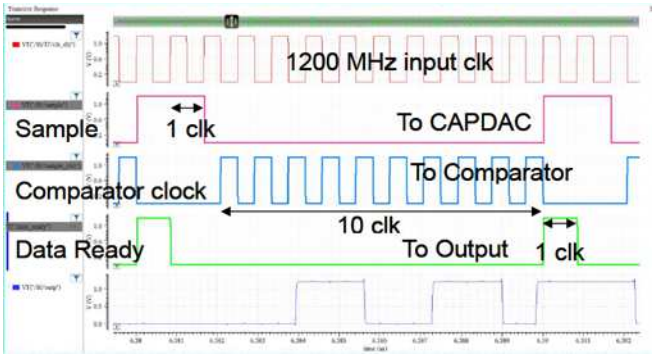


Fig. 6. Timing sequence of the SAR logic

The SAR logic needs exact timing and synchronization to function properly. The sample analog signal must be converted to a digital signal using a total of 12 input clock cycles. To regulate every aspect of the SAR ADC block's logic, the SAR logic will produce three key signals: 1 clock cycle for sampling, 10 clock cycles for bit comparison, and 1 clock cycle for data ready.

This block's operation is coded using Verilog-A. The purpose of this Verilog-A code is to process serial data from an input signal ('comp_out') and produce a variety of output signals. The initialization of internal variables and parameters starts the operation. A control signal ('go') and a clock signal ('clk') are what drive the essential functions.

The code functions as a state-machine, iteratively comparing each input bit ('comp_out') to a threshold ('vth'). Beginning with the most significant bit (MSB), it collects the bits in a methodical manner and moves them from "test_9" to "test_0." Each bit moves between states as it is checked against the threshold, rearranging the bits in accordance.

The Verilog-A line "test_8 = (V(comp_out) > vth)? 1: 0;" demonstrates conditional assignment, comparing the voltage level of "comp_out" to the threshold "vth." It assigns "1" to "test_8" if "comp_out" is greater than "vth," otherwise, it assigns "0," capturing the eighth bit of incoming data.

B. CAPDAC

The capacitive array DAC is an essential component of SAR ADCs, as it generates and subtracts the scaled reference voltage to calculate the digital output. Because capacitance increases exponentially with resolution, the binary weighted capacitor array has constraints in terms of both area and power consumption, even if it offers great linearity. The RC time constants of the driver switches and DAC capacitors are the main reason of the binary-weighted capacitor array's long settling durations. Large power consumption would result from the need for incredibly quick reference voltage buffers in order to charge these massive capacitors at high speeds. Alternative topologies for the DAC are taken into consideration to address these issues.

A further popular substitute is the split binary-weighted capacitive digital-to-analog converter. This topology aids in lowering the array's overall DAC capacitance as well as the distribution of capacitor values. As such, in comparison to the binary-weighted technique, it provides area and power reductions as well as a relaxation of the settling time requirements.

A 10-bit resolution is obtained in this design method, which makes use of the split array DAC depicted in Fig. 7, by combining two binary-weighted capacitive arrays: a 5-bit MSB DAC and a 5-bit LSB DAC. The bridge capacitor C_B, which divides these two arrays apart, is established by:

$$C_B = \frac{Sum\ of\ LSB\ array\ capacitors}{Sum\ of\ MSB\ array\ capacitors} \cdot C = \frac{32}{31}C \qquad (6)$$

To help offset capacitor mismatch faults in the main array of capacitors and enhance the ADC's common-mode rejection, a dummy capacitor is placed at the LSB array.

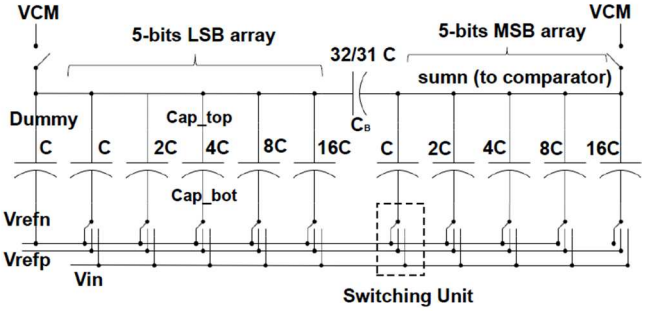


Fig. 7. Split capacitor array DAC (bottom part of differential pair)

With the assumption of a single-ended implementation, find the mismatch-limited minimum value for C_u. Examine the impact of mismatched capacitors in the LSB array DAC, which is diminished by $(1/2^M)$ as a result of the bridge capacitor's existence. In this case, M stands for the MSB array DAC resolution, which is usually selected to be $M \geq N/2$, where N is the ADC's overall resolution. This indicates that the MSB array DAC has a bigger effect on the DAC's overall mismatch performance for larger values of M.

The technique below can be used to determine the worst-case standard deviation of the DNL for the M-bit MSB array DAC in a fully-differential implementation.

$$\sigma_{DNL(max)} = \sqrt{2(2^M - 1)}\left(\frac{\sigma_u}{2C_u}\right) LSB_M \qquad (7)$$

, where C_u and σ_u are the nominal value and standard deviation of the unit capacitor and $LSB_M = \frac{2V_{REF}}{2^M}$.

According to the equation below, this requirement can be written as follows in order to ensure adequate accuracy in the N-bit ADC:

$$3\sigma_{DNL(max)} < \frac{LSB}{2} \qquad (8)$$

, where $LSB = \frac{2V_{REF}}{2^N}$ and combine the two equations previously:

$$\frac{\sigma_u}{C_u} < \frac{1}{3 \cdot 2^{N-M} \cdot \sqrt{2(2^M - 1)}} \qquad (9)$$

For a typical metal capacitor, the standard deviation of capacitor mismatch can be related to the matching coefficient (K_σ), the capacitance density parameter (K_c), and the capacitor area (A) through equation:

$$\sigma\left(\frac{\Delta C}{C}\right) = \frac{K_\sigma}{\sqrt{A}} \ , \ C_u = K_c \cdot A \qquad (10)$$

By combining the previous three equations:

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$$\frac{\sigma_u}{C_u} = \frac{K\sigma\sqrt{K_C}}{\sqrt{2C_u}} = \frac{1}{\sqrt{2}} \cdot \sigma\left(\frac{\Delta C}{C}\right) \quad (11)$$

By substituting this the value of σ_u/C_u will obtaining a mismatch limited lower bound for the unit capacitor which is:

$$C_u \geq 9 \cdot (2^M - 1) \cdot 2^{2(N-M)} \cdot K_\sigma^2 \cdot K_c \quad (12)$$

Given the values $N = 10$, $M = 5$ and matching coefficient K_σ and capacitor density parameter K_c from the design kit documentation which is $3.94\%.\mu m$ and $0.011\ fF/\mu m^2$. The minimum value of C_u is $5fF$.

Nevertheless, the computed value of C_u will be somewhat greater due to practical limitations like the layout parasitic and the minimum capacitor value specified by the design kit. This number offers a sensible compromise between practical implementation considerations and accuracy.

As seen in Fig. 8, the differential capacitive DAC uses two split capacitor arrays. In addition, CMOS transmission gates are the switches that the DAC uses to switch the VCM during the sampling phase. A switching unit, as seen in Fig. 9, is utilized to switch between the sampled input and the DAC bit selection.

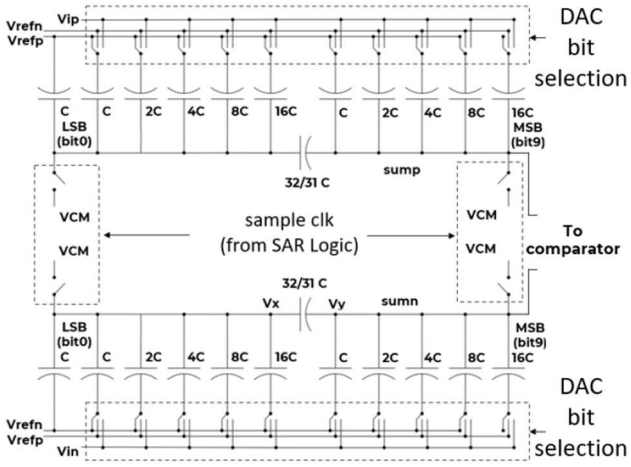


Fig. 8. Proposed differential split capacitor array DAC

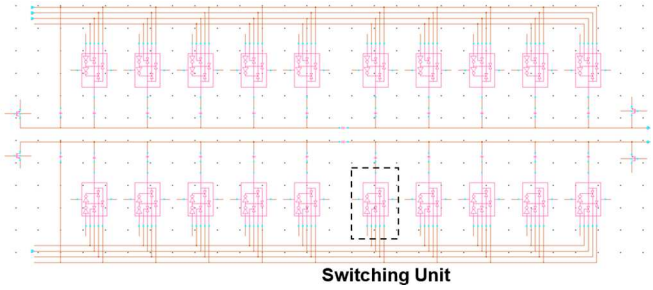


Fig. 9. Schematic of differential split capacitor array DAC

C. Switching Unit

Fig. 10 shows the switching unit employing transistor-level implementation in this suggested architecture.

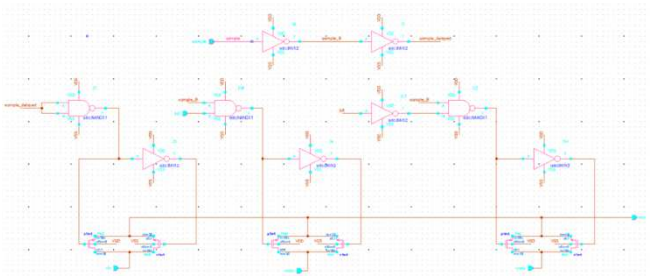


Fig. 10. Schematic of switching unit

The CMOS transmission gate is used to implement the switch. Compared to bootstrapped switches, CMOS transmission gates can be more area-efficient and easier to build. When appropriately scaled, the switches offer low ON resistance and negligible signal distortion. The ON resistance impact and signal distortion can be further reduced by adjusting the multiplier and fin counts in the transistor. Additionally, by simultaneously adding the PMOS and NMOS FinFET. Because both operate with two opposing clocks (sample_delayed & sample_delayed_b), the clock feedthrough effect is eliminated.

The largest capacitor in this switching unit arrangement, 16C, dictates the size of the transistor. As a result, the multiplier count is fixed at 8, and the number of fins is fixed at 6. Given that the switching unit outputs to the capacitor, these values are determined based on the settling time needed for the voltage of the capacitor. As a result, the capacitor's charging time must be carefully considered, and speed is required because of the high sampling frequency. Altering the transistor's fin count and multiplier count will enhance drive current, resulting in faster switching and less delay. By looking at the capacitor settling waveform, the values are found.

D. Dynamic Comparator

It is because of dynamic comparators can reduce static bias currents and increase power efficiency; these designs are frequently utilized in SAR ADC implementations. The reset phase and the evaluation phase are the two stages of the clock signal-based dynamic comparator's operation. A dynamic comparator's primary characteristics include kickback noise, offset voltage, resolution, and gain.

The comparator design comprises of an RS flip-flop after a StrongArm latch. StrongArm comparators include a regenerative latch that allows to run at low voltage and provide full rail-to-rail output swing. This design also consumes almost little DC current, which results in low power consumption. A clock signal activates the StrongArm latch, which then transforms the differential signal applied at the input into a 1-bit digital code. Either GND or VDD are the output levels. The differential voltage is swiftly converted into digital levels by the StrongArm latch using regenerative gain. The RS flip-flop converts the latch's output into digital levels and adds further amplification. In order to guarantee that the signals are compared (comparator operates) when the input signals are stable, a clock driver is included.

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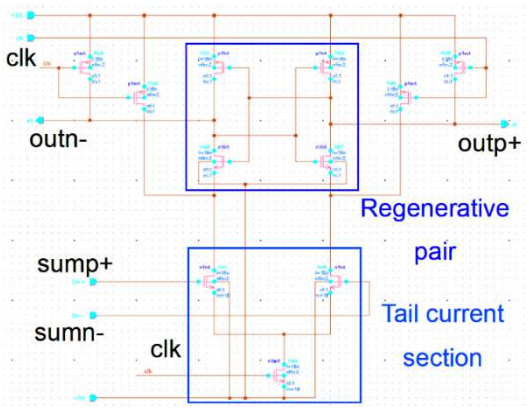


Fig. 11. Schematic of StrongArm latch

The reset phase and the evaluation/regenerative phase are the two stages of the single-stage dynamic comparator's functioning, which primarily focuses on the StrongArm latch as seen in Fig. 11. When the "clk" signal is low during the reset phase, the prior analog input is cleared, PMOS turns on while NMOS turns off, and the output is charged via the MOSFET loading capacitor to the supply voltage (VDD). In this phase, there is no static power consumption due to no current draw. During the regeneration phase, the cross-coupled inverter section (regenerative pair) provides positive feedback that generates the amplification output when the "clk" signal is high. It allows the translation of minor input differences to full-scale digital levels because the output that is produced depends on the differential input.

The transistor (MN1), whose gate signal is $sump+$, will draw more current in its branch during operation if the voltage at $sump+$ is higher than $sumn-$. This will cause the $outn-$ to discharge more quickly and pull towards ground. As the output of one inverter is the input of another, turn on the transistor (PM5) whose gate signal is $outn-$ first. As a result, as Fig. 11 illustrates, $outp+$ will pull to Vdd and $outn-$ will regenerate to ground.

Due to the requirement for a high tail current strength in this switching, the comparator block's tail current section must have a minimum of 4 multipliers for the transistor. By improving the charging and discharging capacities of the comparator's internal nodes during the switching process, this increase in current results in faster switching and shorter propagation delays. The comparator's speed, offset, and noise performance are all significantly influenced by the tail current. A higher tail current can increase the comparator's speed. On the other hand, an excessive tail current may result in needless heat dissipation and higher power usage. As a result, counting the multipliers on the transistor depends on keeping an eye on the comparator's output. In the event that the waveform does not settle during the evaluation phase, it will be essential to increase the number of multipliers.

IV. Circuit Design

A. Transient Analysis

The ADC (analog) output will show up in the ViVA window following the transient simulation's conclusion, and FFT will be performed on it as seen in Fig. 12.

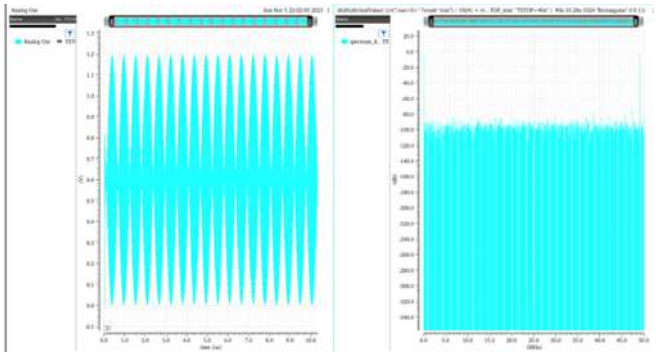


Fig. 12. Analog output and FFT of ADC

The dynamic measurement results are then obtained utilizing the ViVA Spectrum measurement, as indicated in Fig. 13.

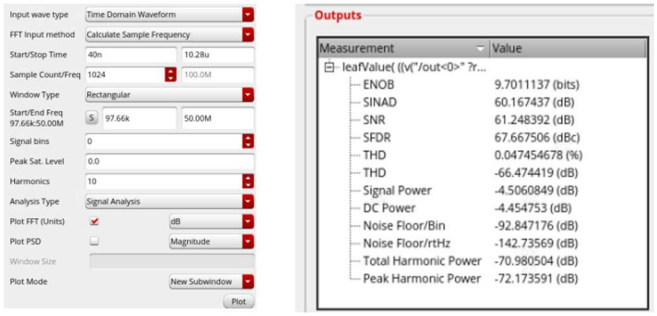


Fig. 13. ADC Transient Analysis Results

B. Overall Output Results

Fig. 14 displays the testbench used to test the overall SAR ADC. Fig. 15 plots all the significant node voltages from the testbench. The testbench generates an output of 100 MSPS using an external clock of 1200 MHz, two inputs, "vip" and "vin," each with a frequency of 110 kHz, $\frac{11}{120000} \times 1200MHz$, and an amplitude of 0.5 V (ranging from 0.36 V to 0.86 V.

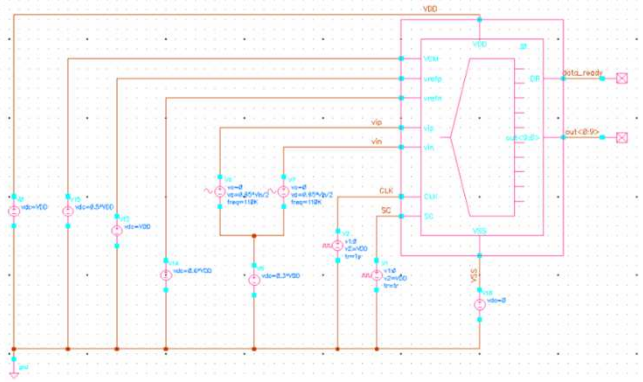


Fig. 14. Testbench for analag output and FFT of ADC

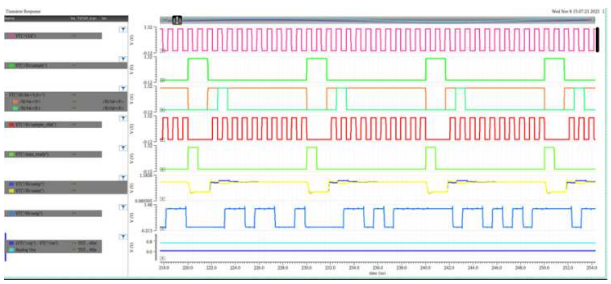


Fig. 15. Transient Waveform of the SAR ADC testbench

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When the external clock signal "CLK" rises, the "sample" clock signal, which regulates the input signal sampling, is generated. the "sample" clock signal following twelve cycles of the "CLK" external clock signal. When "sample" drops below a certain threshold, the SAR logic block's control bit "bit<9>" rises to a certain level, redistributing the charges in the capacitive DAC that controls the switching unit and altering the voltage of the comparator's "sump" and "sumn" input nodes. The comparator is triggered by the rising edge of the "sample_clkb" and the DAC would have already settled by the halfway point of the "bit<9>."

By the halfway point of the "sample_clkb," "high" or "low" was already generated by the comparator output. The SAR logic then begins to run. When the "sample_clkb" falls edge, "bit<9>" will continue to switch high and "bit<8>" will switch to high if the comparator output is high. Conversely, "bit<9>" will switch low and "bit<8>" will switch to high if the comparator output is low. This process is repeated until "bit<0>," which occurs ten cycles later for "sample_clkb." The "data_ready" signal will then go high, indicating that the SAR logic block has produced the output digital code.

Table I and Table II present the proposed performance summary and performance comparison, respectively. Three literature results have been used in comparison. From the comparison table, it is evident that the proposed ADC demonstrates competitive performance compared to the references, especially in terms of resolution, power consumption, and Figure of Merit (FOM), despite utilizing a more advanced 18nm FinFET CMOS technology. The proposed ADC achieves a high Effective Number of Bits (ENOB) and Signal-to-Noise and Distortion (SINAD), coupled with low total power consumption. These attributes make it well-suited for applications requiring high-resolution and high-speed analog-to-digital conversion.

TABLE I. PROPOSED ADC PERFORMANCE SUMMARY

Parameter	Value
Technology	18nm FinFET CMOS
Power Supply	1.2 V
Resolution	10 bits
Sampling Rate	100 MSPS
Full Scale Input	0.96 Vpp
ENOB	9.7 bits
SINAD	60.17 dB
SNR	61.25 dB
SFDR	67.67 dBc
Conversion Time	12 clock cycles
Total Power	0.216 mW
FOM	2.6 fJ/conversion-step

TABLE II. PERFORMANCE COMPARISON

Reference	[9]	[10]	[11]	This work
Technology	65nm	90nm	40nm	18nm
Supply Voltage (V)	1.2	1.2	0.9	1.2
Resolution (bit)	11	10	12	10
Conversion Rate (MS/s)	100	50	100	100
Input Range (V)	2.0 Vpp	2.0 Vpp	1.8 Vpp	0.96 Vpp
DNL (LSB)	1.28	0.36	1.13	1
INL (LSB)	1.05	0.45	0.93	1
ENOB (bits)	9.51 @49.23MHz	9.26 @5.0MHz	10.85 @2.11MHz	9.7 @49.12MHz
Power (mW)	1.6	0.664	2.6	0.216
FOM (fJ/Conv.-step)	21.9	21.68	14.6	2.6

V. CONCLUSION

This paper presents the developed and simulated 10-bit asynchronous SAR ADC that utilizes the 18nm FinFET CMOS technology, running at 1.2 V nominal voltage. With an internal clock generator that synchronizes the output of the clock signal to the SAR logic, the ADC architecture orchestrates a smooth conversion process, allowing for an impressive 100 MSPS sampling rate versus a 1.2 GHz clock input.

The research on this topic has demonstrated the effectiveness of utilizing a fully differential pair capacitor array to achieve high accuracy and resolution in ADCs. This proposed design offers a simplified IC layout, while the incorporation of digital control through SAR logic implemented in Verilog significantly reduces the complexity of analog circuit design complexity and requirement. Future research could focus on ways of improving the proposed design intended for higher sampling rate applications.

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